

Homework Assignment 2

Instructions. Submit your answers as a PDF on Webcourses by **September 25 at 11:59 p.m.** I encourage you to work in groups and discuss the problems with one another. You may also use AI tools, but remember that AI will not be available during exams. Use these problems to check your own understanding of the material. Your grade is based on participation, not correctness. You must submit your answers to receive credit for this assignment.

Justify your answers. In each induction proof, state the claim, check the base case or cases, state the induction hypothesis, and prove the inductive step. Throughout this assignment, $\mathbb{N} = \{0, 1, 2, \dots\}$.

Question 1 Set notation and set identities

Let $A = \{\emptyset, \{1\}, \{1, 2\}\}$. Recall that $\mathcal{P}(A)$ is the set of all subsets of A .

(a) Determine whether each statement is true or false, and briefly explain each answer:

- (i) $\emptyset \in A$ (ii) $\emptyset \subseteq A$ (iii) $\{\emptyset\} \subseteq A$,
 (iv) $\{1\} \subseteq A$ (v) $\{1\} \in A$ (vi) $\{\{1, 2\}\} \in \mathcal{P}(A)$.

(b) Find $|A|$ and $|\mathcal{P}(A)|$. Explain why the nested braces matter.

(c) For arbitrary sets X, Y, Z , prove the identity

$$X \setminus (Y \cup Z) = (X \setminus Y) \cap (X \setminus Z).$$

Use an arbitrary element and logical equivalences, or prove both inclusions. A diagram alone is not a proof.

Question 2 Counting disjoint pairs of subsets

Let S be a set with n elements. Consider ordered pairs (A, B) such that

$$A \subseteq S, \quad B \subseteq S, \quad A \cap B = \emptyset.$$

Either subset may be empty. Order matters: (A, B) and (B, A) count as different pairs whenever $A \neq B$.

Prove by induction on $n \geq 0$ that there are exactly 3^n such pairs.

Hint. Remove one element x from S . Consider the three possibilities for where x can appear in a pair (A, B) .

Question 3 Countably infinite sets

An infinite set is *countably infinite* if its elements can be listed with indices $0, 1, 2, \dots$ so that every element appears exactly once, at a finite index. Consider

$$M = \{0, 3, 6, 9, \dots\}, \quad R = \{1, 2, 4, 5, 7, 8, \dots\},$$

where M consists of the nonnegative multiples of 3, and R consists of the positive integers that are not multiples of 3.

- (a) Give a rule that pairs each $n \in \mathbb{N}$ with an element of M . Write the first six pairings and explain why every element of M is paired with exactly one natural number.
- (b) List the first twelve elements of R in increasing order. Explain why continuing this pattern lists every element of R exactly once at a finite index, so R is countably infinite.
Hint. Every element of R has exactly one of the forms $3k + 1$ or $3k + 2$, where $k \in \mathbb{N}$.
- (c) A student claims that M has smaller cardinality than $\mathbb{N}$ because M is a proper subset of $\mathbb{N}$. Is this argument valid? Explain using your pairing from part (a).

Question 4 Ordinary induction

Prove each statement by ordinary induction.

- (a) For every integer $n \geq 0$,

$$\sum_{k=1}^n (2k - 1) = n^2.$$

The sum is defined to be 0 when $n = 0$.

- (b) For every integer $n \geq 0$,

$$8 \mid (3^{2n} - 1).$$

Recall that $8 \mid t$ means $t = 8q$ for some integer q .

- (c) For every integer $n \geq 0$,

$$\sum_{k=0}^n 2^k = 2^{n+1} - 1.$$

- (d) For every integer $n \geq 4$,

$$2^n \geq n^2.$$

- (e) Define a sequence by $a_0 = 1$ and $a_{n+1} = 3a_n + 2$ for every integer $n \geq 0$. Prove that

$$a_n = 2 \cdot 3^n - 1 \quad \text{for every integer } n \geq 0.$$

Question 5 Counting with inclusion–exclusion

A survey asks 200 students which programming languages they have used. Let P , J , and C be the sets of students who have used Python, Java, and C++, respectively. The survey reports

$$\begin{aligned} |P| &= 110, & |J| &= 85, & |C| &= 70, \\ |P \cap J| &= 45, & |P \cap C| &= 35, & |J \cap C| &= 30, \\ |P \cap J \cap C| &= 15. \end{aligned}$$

Each pairwise count includes the students who have used all three languages. Show your calculations; you may draw your own Venn diagram to organize the information.

- How many students have used exactly one of these three languages?
- How many students have used exactly two of these three languages?
- How many students have used at least one of these three languages? How many have used none of them?

Question 6 Understanding a halving algorithm

The following algorithm takes a positive integer n as input. The expression $\lfloor x/2 \rfloor$ means divide by 2 and round down; for example, $\lfloor 5/2 \rfloor = 2$.

```
HalvingCount(n):
    x = n
    count = 0
    while x > 1:
        x = floor(x / 2)
        count = count + 1
    return count
```

- Trace the algorithm for $n = 20$. Record x and `count` initially and after each loop iteration. What is returned, and why does the loop stop?
- What is returned for $n = 1, 8, 16$, and 32 ? More generally, what is returned when $n = 2^k$ for an integer $k \geq 0$? Explain the pattern.
- Restrict attention to $n = 2^k$, where $k \geq 0$ is an integer. Consider two sequences with $T_1(1) = T_2(1) = 0$ and, for $n \geq 2$,

$$T_1(n) = T_1(n/2) + 1, \quad T_2(n) = 2T_2(n/2) + 1.$$

Find an exact formula for each sequence in terms of n , and prove both formulas by induction on k . Compare their rates of growth. The first sequence counts iterations of the halving algorithm; the second is a separate recurrence.

Hint. Compute the values of both sequences at $n = 1, 2, 4, 8$ before guessing a formula.

Question 7 Proving selection sort correct by induction

Selection sort repeatedly finds a smallest entry in the remaining part of an array and moves it into the next position. Consider this algorithm for an array $A[1], \dots, A[n]$ of integers, where $n \geq 1$. Repeated values are allowed. The goal is to put the entries in *nondecreasing order*: $A[1] \leq A[2] \leq \dots \leq A[n]$.

```

SelectionSort(A):
    n = len(A)
    for i = 1, ..., n - 1:
        m = i
        for j = i + 1, ..., n:
            if A[j] < A[m]:
                m = j
        swap A[i] and A[m]
    return A

```

Swapping a position with itself leaves the array unchanged. When $n = 1$, the outer loop has no iterations.

- (a) Trace the algorithm on

$$A = [5, 2, 4, 2, 1].$$

For each outer-loop iteration, record i , the chosen index m , and the entire array after the swap. Include iterations that leave the array unchanged.

- (b) State and prove an induction claim about the array after k completed outer-loop iterations, for $0 \leq k \leq n - 1$. Your claim should describe:

- the order of the first k entries;
- how those entries compare with the remaining entries; and
- why every original value still occurs the same number of times.

Use $k = 0$ as your base case. In the inductive step, justify why the inner loop finds a minimum among the remaining entries and why placing it in the next position preserves your claim.

- (c) Use your induction result to prove that selection sort terminates and returns the original entries in nondecreasing order. Explain why your argument also covers $n = 1$ and arrays with repeated values.